CS 7545 F26, Computational Mini Project 2

100 point assignment

Due September 24, 2026

In this project, you'll visualize and compare generalization error and uniform convergence for finite hypothesis classes.

Let $\mathcal{X}$ be a domain and D be a distribution over $\mathcal{X} \times \{-1, 1\}$.

Let $H \subset \{h : \mathcal{X} \to \{-1, 1\}\}$ be a hypothesis class.

Define:

$$L(h) = \mathbb{E}_{(x,y)\sim D}[\mathbf{1}[h(x) \neq y]] \tag{1}$$

$$\hat{L}_S(h) = \frac{1}{|S|} \sum_{(x,y)\in S} \mathbf{1}[h(x) \neq y] \tag{2}$$

$$h^\star = \arg\min_{h\in H} L(h) \tag{3}$$

$$\hat{h}_S = \arg\min_{h\in H} \hat{L}_S(h) \tag{4}$$

Part 1 (10 points)

Run the provided code to visualize the distribution of D formed by 3 Gaussian distributions. The code displays the distribution as an array of dots, where the size of the dot at (x_1, x_2) corresponds to $\Pr(X = (x_1, x_2))$ and the color of the dot corresponds to $\Pr(Y = 1|X = (x_1, x_2))$, from blue at 0, white at 1/2, and red at 1.

Sample a subset S of size 100 using `seed=0` then use the provided code to visualize the set S. The positive points are shown in red and the negative points are shown in blue. When multiple points are sampled at a location, they are arranged in a small circle.

Part 2.a (20 points)

Sample a subset S of size 25 using the first 5 seeds and compute $\hat{h}_S$ and $h^\star$.

Run the code to visualize $\hat{h}_S$ and $h^\star$ against both the distribution D and the set S. Additionally, calculate $L(h^\star)$, $L(\hat{h}_S)$, $\hat{L}_S(h^\star)$, and $\hat{L}_S(\hat{h}_S)$.

In your report, include the loss numbers, the distribution plot and the dataset plot for a single seed (and include which seed). Additionally, include any plots or numbers to discuss the following questions:

- How does $\hat{h}_S$ vary with S?
- What are the relationships between the loss numbers, and why?

Part 2.b (10 points)

Repeat the experiment from Part 2.a, but reverse the ordering of H to prefer smaller boxes that minimize the empirical loss, rather than larger boxes.

In your report, include the loss numbers, the distribution plot and the dataset plot for a single seed (and include which seed). Additionally, include any plots or numbers to discuss the following question:

- How does $\hat{h}_S$ change compared to Part 2.a?

Part 3.a (10 points)

Fill in `def Get_h_max(dataset, distribution, hypothesis_class):` to return the hypothesis $h_{\max}$ that maximizes $L(h) - \hat{L}_S(h)$. Include the code snippet in your report.

Part 3.b (20 points)

For each of the first 10 seeds, plot a histogram of $L(h) - \hat{L}_S(h)$ and mark a line for the value of $h^\star$, $\hat{h}_S$, and $h_{\max}$.

In your report, include the histogram for a single seed (and include which seed). Additionally, include any plots or numbers to discuss the following questions:

- What is the general shape of the histogram?
- How does the shape of the histogram change for different seeds and what is a brief explanation?
- How does the distribution of $L(h^\star) - \hat{L}_S(h^\star)$ behave?
- How does the distribution of $L(\hat{h}_S) - \hat{L}_S(\hat{h}_S)$ behave?

Part 4 (30 points)

Recall from class that for finite hypothesis classes, with probability at least $1 - \delta$,

$$\max_{h \in H} L(h) - \hat{L}_S(h) \leq \sqrt{\frac{\ln(|H|) + \ln(1/\delta)}{2n}} \tag{5}$$

Calculate and report these two quantities using $\delta = 0.05$, the H used in the previous parts, and both $n = 25$ and $n = 100$. Also, for both values of n, compute the value of the LHS for the first 5 seeds.

- Briefly discuss how the bound compares to the empirical values.
- Discuss the effect of n on both the bound and the empirical values.